Proof. (i) Let V be an irreducible representation of A. Let $v \in V$. Then $Iv \subset V$ is a subrepresentation. If $Iv \neq 0$, then $Iv = V$ so there is $x \in I$ such that $xv = v$. Then $x^n \neq 0$, a contradiction. Thus $Iv = 0$, so I acts by 0 in V and hence $I \subset \operatorname{Rad}(A)$.

(ii) Let $0 = A_0 \subset A_1 \subset \cdots \subset A_n = A$ be a filtration of the regular representation of A by subrepresentations such that A_{i+1}/A_i are irreducible. It exists by Lemma 3.4.2. Let $x \in \operatorname{Rad}(A)$. Then x acts on A_{i+1}/A_i by zero, so x maps A_{i+1} to A_i. This implies that $\operatorname{Rad}(A)^n = 0$, as desired. □

Theorem 3.5.4. *A finite dimensional algebra A has only finitely many irreducible representations V_i up to an isomorphism. These representations are finite dimensional, and*

$$A/\operatorname{Rad}(A) \cong \bigoplus_i \operatorname{End} V_i.$$

Proof. First, for any irreducible representation V of A and for any nonzero $v \in V$, $Av \subseteq V$ is a finite dimensional subrepresentation of V. (It is finite dimensional as A is finite dimensional.) As V is irreducible and $Av \neq 0$, $V = Av$ and V is finite dimensional.

Next, suppose we have nonisomorphic irreducible representations $V_1, V_2, \ldots, V_r$. By Theorem 3.2.2, the homomorphism

$$\bigoplus_i \rho_i : A \longrightarrow \bigoplus_i \operatorname{End} V_i$$

is surjective. So $r \leq \sum_i \dim \operatorname{End} V_i \leq \dim A$. Thus, A has only finitely many nonisomorphic irreducible representations (not more than $\dim A$).

Now, let $V_1, V_2, \ldots, V_r$ be all nonisomorphic irreducible finite dimensional representations of A. By Theorem 3.2.2, the homomorphism

$$\bigoplus_i \rho_i : A \longrightarrow \bigoplus_i \operatorname{End} V_i$$

is surjective. The kernel of this map, by definition, is exactly $\operatorname{Rad}(A)$. □

Corollary 3.5.5. $\sum_i (\dim V_i)^2 \leq \dim A$, *where the V_i's are the irreducible representations of A.*